

Medibot-Healthcare Robot Assistant for Remote Communities

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Abstract—Across the world, the lack of proper healthcare resources in rural areas significantly impacts the physical and economic well-being of individuals living there. There is a need for multi-faceted solutions to help alleviate healthcare issues in these areas. Medibot is a robot which monitors the health of individuals in these areas and takes appropriate measures. Patients visit Medibot for daily non-invasive monitoring of their vitals and then receive updated guidance or are connected with a healthcare professional for immediate support. The robot provides an easy way to get continuous care for those with limited access to healthcare resources or facilities.

Index Terms—Human-Robot Interaction; Medical Assistant; Socially Assistive Robots; Healthcare

I. INTRODUCTION

Access to healthcare in remote areas is more difficult and challenging compared to urban areas. There are many issues in creating reliable access to healthcare in rural areas, including higher costs to deliver supplies and equipment, lack of trained professionals, limited digital access, financial barriers, and other socioeconomic factors [1]. With limited supplies and medical access in these vast areas, it can mean that patients cannot afford to purchase supplies and may need to travel long distances to speak to someone about their concerns [2]. Patients in these areas often face many socioeconomic barriers to obtain the care they need, such as facing stigma from healthcare professionals in the community and limited knowledge or confidence in the services provided [3]. These complex problems require innovative solutions that combine policy, technology and urban planning to ensure those living in remote areas have proper access to healthcare. To address these concerns, we have designed a social robot called Medibot, a healthcare assistant that can be placed in central locations of remote communities (e.g. local town hall, community center) to assist patients in non-invasive monitoring, receiving guidance for their health concerns and connection with healthcare professionals for immediate support. This data-driven solution will create immediate healthcare access for patients and provide a friendly simplified interaction for patients. The robot provides consistent and constant support for locals living in these communities as they can use the tools provided to monitor any existing health conditions or test for any areas of concern. Patients may feel more comfortable connecting

with a digital assistant who will provide advice based on the individual's profile.

II. UNITED NATIONS SUSTAINABLE DEVELOPMENT GOALS

The development of this robot was inspired by Sustainable Development Goal (SDG) 3: Good Health and Well-Being and SDG 10: Reduced Inequalities [4], [5]. Ensuring proper access to healthcare resources and facilities is crucial to fostering a healthy and prosperous population which can thus create a resilient economy [5]. Many individuals who live in rural areas around the world do not have access to robust healthcare systems and services which can be further exacerbated by inequalities due to race, gender and other factors [6]. Ensuring the presence of healthcare facilities in rural areas can be difficult due to factors such as a lack of healthcare practitioners in the region, high transport costs and lack of funds and resources to support the infrastructure. Innovative approaches are required to circumvent these issues and provide quality health care to patients living in places which may be more disadvantaged in getting medical treatment. Robotics can play a pivotal role in providing extra assistance in these areas.

III. DESIGN

We started investigating this problem through research and identifying different needs and requirements based on the experiences of those working in remote areas. We spoke to a doctor currently working in a remote area of a developing country who said that while limited resources in a country can cause illnesses to become more critical, distance is the main reason why individuals don't get the appropriate treatment earlier. Based on our conversation and research we identified the following areas. Our main focus for this project was the experience of the user as they completed the human-robot interaction.

A. Remote Location

Many patients in remote areas live far away from their local clinic or medical establishment. These remote areas may also have limited access to electricity or technology. Instead of using a pre-existing social robot, which can be resource-heavy, we designed a prototype that is modular and

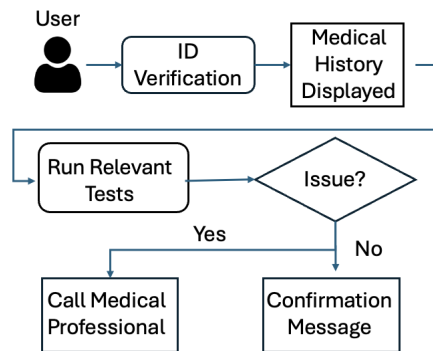


Fig. 1: User Interaction Flow Diagram

lightweight. Personal robots currently have a high cost which can be a financial load on the healthcare system and on patients. We decided that the Medibot will be a communal robot placed in a central building which can be regularly accessed by residents. This will greatly reduce travel distances to get healthcare advice, time required to visit and potentially incentivize patients to get help sooner rather than later.

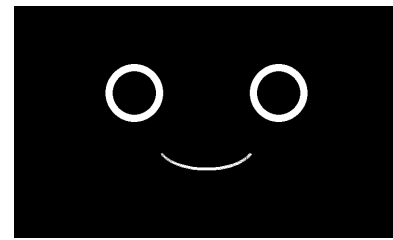
B. Medical Expertise

Patients need appropriate healthcare support to address preliminary concerns and monitor low risk long-term illnesses. However, it is also important to note that depending on the severity of the issue, it is preferred to have a medical professional interact with the user. With Medibot, patients in the region can check their vitals and be monitored for long-term illnesses on a daily basis or as needed by a physician who may not be available to visit often. Medibot leads patients through the entire interaction as seen in Figure 1. Patients in the area will approach the robot and provide their identification. After verifying their identification and based on their medical history, Medibot will lead them through a series of tests either to ensure compliance with medical treatment or to monitor for potential diseases. Once the tests are complete, the robot will record this data and connect the patient to a healthcare provider if there are any major immediate concerns to address or it will provide a confirmation to the patient with a reminder of the next visit. It is important to note that Medibot is an assistant and thus if there are any severe issues, would direct to a medical professional immediately.

C. Individual Preferences

The robot needs to be friendly and easy to use, guiding the user through the different steps of the potentially intimidating interaction. It is important that the robot does not intimidate the user or add additional stress and encourages the user to use the system in the future. We addressed this problem using two specific methods: facial expressions and light signals.

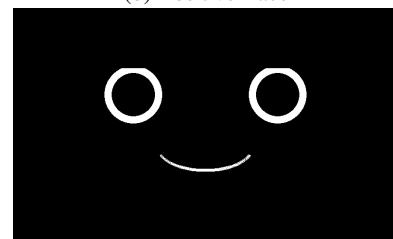
1) *Facial Expressions*: Facial expressions are important for social robots as they display an identity which draws users to form a human-robot interaction. Mechanical faces can be



(a) Neutral Face



(b) Positive Face



(c) Worried Face

Fig. 2: Facial Expressions of the Medibot (a) Neutral Face (b) Positive Face (c) Concerned Face

difficult to implement and can be jarring for users so we utilized a digital screen to project a simple face [7], [8]. Based on previous research we used a rectangular face to avoid specific gender stereotypes and included some cheeks as it was deemed as most human-like, friendly, intelligent and trustworthy [9]. Additional research showcases that if the features are proportionately in the middle there is an increased level of trustworthiness which was also implemented on the screen [10]. Figure 2 showcases the three faces that we used for the robot design.

We implemented three facial expressions to represent different behaviours as the user interacts with the robot. The first face is a neutral face as seen in Figure 2a. We kept the design simple and still welcoming to the user. A more positive face, as shown in Figure 2b is used when something has been a success (e.g. inputting data, getting appropriate test results). Finally, the worried face as shown in Figure 2c is used for when there might be something wrong and human intervention might be required. We wanted it to be a lightly worried face instead of something that is very concerned because patients might be already apprehensive about their check-ups or interactions and we did not want to add additional stress to the user during the interaction.

2) *Light Behaviors*: Previous research has shown that the addition of lights can help participants understand the intent of a robot and provide context for the current state [11],

[12]. To help users understand the state and motion of the robot, the LEDs had multiple colours. We combined the lights with the facial expressions to make a cohesive and intuitive interaction. A yellow light showcased a concern such as issues with results, or where a medical professional needed to be called. The reason we didn't use a red light which is typically used as a warning symbol was similar to why the worried face wasn't extreme as patients may get concerned if they see the color red which is often associated with danger. A green light showcased that things were going well and is often paired with the positive facial expression but also marks the completion of tasks. The blue light showcased that the robot was processing or that it required input from the user. The single flashing blue light would also appear on one arm if the user needed to utilize a specific device located on it. The lights can be paired with different facial expressions however for cohesive behaviors the yellow light is often paired with the worried facial expression while the blue and green light are interchangeable for the neutral face.

IV. IMPLEMENTATION

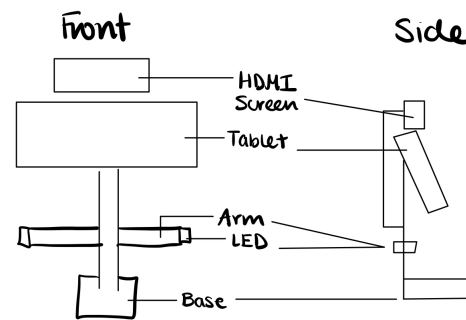
Medibot is a stationary social robot with a screen for some facial expressions, a tablet for interactivity and limbs to hold medical devices that will be used by the patients to monitor their vitals. The robot is meant to be a friendly alternative to a healthcare professional who may not be available who can complete initial screening and testing of the user.

A. Mechanical Design

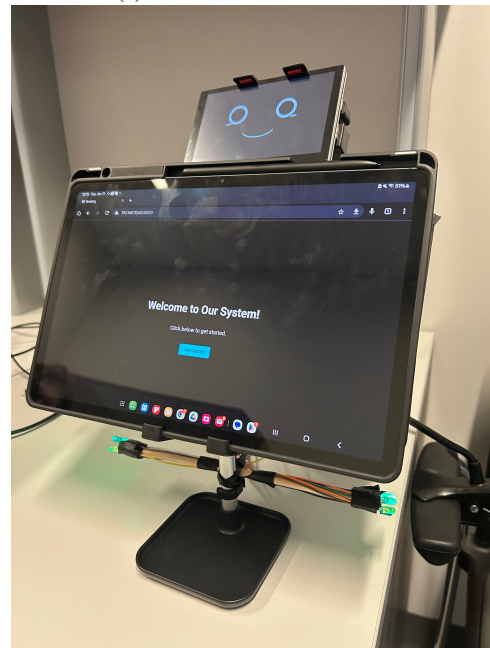
We sketched the design for the robotic frame and then decided to create the frame for the Medibot prototype using different scrap mechanical components. We chose to avoid using a pre-existing social robot as many are expensive and their capabilities did not align with the needs of this robotic design. We use a stand for the tablet as the body of the robot which is the initial structure of the robot and frame for all the remaining components. A horizontal rod was added to the frame as an arm to hold various devices perpendicular to the main frame. The LED lights are placed at the ends to signal to the user. A smaller screen was attached at the top to show the facial expressions. We decided to include a smaller screen at the top for facial expressions because the main screen seemed quite large and the separation of the two features ensured that the information was kept away from the robot's embodiment. Figure 3 showcases the initial design and the final prototype for this robot.

B. Electrical Components

Medibot utilizes a simple electrical system centered around our single-board computer, a Raspberry Pi 3 Model B. An HDMI screen is connected for facial expressions for interactions and LEDs for signalling. Further information about each section can be found below.



(a) Initial Medibot Sketches



(b) Current Medibot Prototype

Fig. 3: Design Process for Medibot

1) *Interactive Tablet and HDMI Screen:* We utilized an HDMI Screen to incorporate facial expressions. Since the social robot will deliver information to the user, we wanted to include a method for the user to get information visually and simplify some potentially difficult feedback that the robot would provide. On the tablet screen, the user can input their information and see all the instructions, metrics and advice. The visual aid allows the social robot to demonstrate different actions without needing a physical embodiment which is often difficult to embed within a social robot due to the limited capabilities and power requirements. Figure 4 shows a sample screen the user may see.

2) *LED Lights:* We utilized very simple LED lights connected to the Raspberry Pi through resistors. Using these connections, we were able to control and synchronize the light signals with the state of the system. Blue, green, and yellow lights were used. The green and yellow lights were paired and could only be turned on on both sides of the arm or completely turned off. However, the blue lights could be

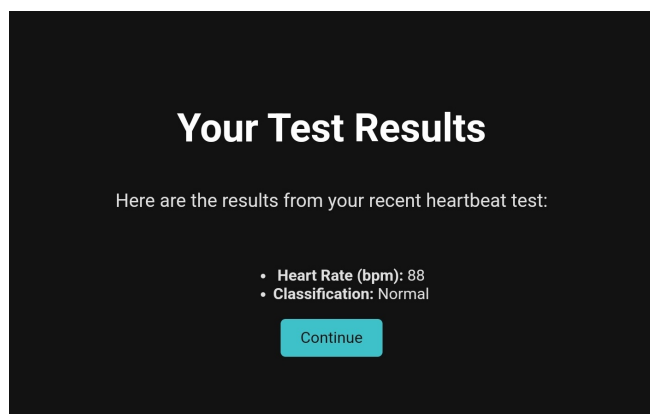


Fig. 4: Test Results Screen

individually controlled on both sides of the arm so that they could be used to signal to the user that they needed to interact with something on that side of the arm.

V. CONCLUSION AND FUTURE WORK

This project introduces a prototype of a social robot to be used as a medical assistant and monitoring device in rural areas. Our design was based on research about limitations in rural areas which causes a lack of medical resources. The prototype demonstrates a new way of developing innovative multi-modal systems to support the burdened healthcare systems in those areas. This tool may be an effective alternative to alleviate some of the burdens faced by rural communities when finding consistent medical services for daily issues and providing more proactive medical services.

In the future, we would like to embed more co-design approaches. By connecting with experienced healthcare professionals who have worked in these settings and relevant stakeholders with lived experiences, we will get more knowledge and personal anecdotes to further shape the robot design. These interactions will allow us to iterate and build a more reliable and relevant robotic system.

The integration of medical monitoring devices needs to be further explored. The focus of this iteration was the social behaviours of the robot. Current non-invasive technologies that can monitor different vitals and connect to an API are relatively expensive, thus, it was not feasible to integrate them into the design of this iteration. Another area which would be important to explore are guidelines around data privacy for this medical information. There are strict guidelines across the world regarding testing and to ensure the safety of patient's data it is important that these guidelines and laws are taken into consideration when developing this robot.

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